1. Finance of Supply Chain Management

Everything is **stocks** and **flows**

1.1. Accounting Rules

1. Financial (transactions -> accounts)
    - Provides information primarily to people **outside** the company
        - Equity & Debt
        - Credit from suppliers
        - Customers
        - Employees
2. Managerial
    - Provides information primarily to people **inside** the company
        - internal investment decisions
        - performance monitoring & evaluation

1.2. Concepts of Financial Accounting

- Conservatism:
- Materiality:
- Consistency:
- Economic Entity:
- Going Concern:
- Accounting Period:
- Cost Concept:

1.3. Principles of Accrual Accounting

Attempt to measure firm performance **regarless of when cash is exchanged**

- Revenue Recognition
    - Earnings process complete
    - Cash collection assured
- Mathcing Principles
    - Match effort to the benefits generted
    - Capitalize expenditure that will benefit future periods, expense as benefit are realized
    - Recognize liability when efforts benefiting the period require cash payment in the future

1.4. Financial Statements

1.4.1. Balance Sheet (BS)

$$\underbrace{\text{Assets}}_{\text{own}} = \underbrace{\text{Liabilities}}_{\text{owe}} + \underbrace{\text{Equity}}_{\text{shareholders}}$$

$$\text{Equity} = \text{Assets} - \text{Liabilities}$$

Assets	Liabilities
Cash	Account Payables
Account Receivables	Debt
Inventory	Equity
PP&E	

- **Cash**: Money immediately available to spend.

  $50,000 sitting in the company's checking account.

- **Accounts Receivable**: Money owed by customers for sales already made on credit.

  You deliver $10,000 of consulting work in March, invoice the client with 30-day terms — that $10,000 is receivable until they pay.

- **Inventory**: Goods held for sale but not yet sold.

  A shoe retailer has $200,000 of unsold shoes sitting in its warehouse.

- **PP&E**: Long-lived physical assets used to run the business.

  A factory building and its production machinery, carried at $2M after depreciation.

- **Accounts Payable**: Money owed to suppliers for goods/services already received but not yet paid for.

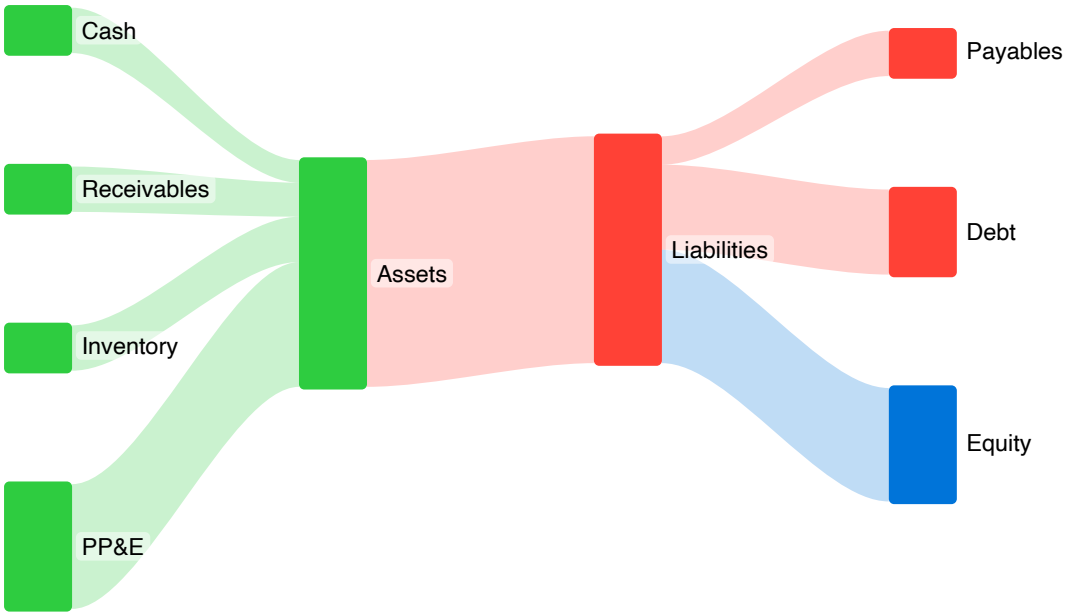
  You receive $5,000 of raw materials from a supplier on 60-day payment terms — that $5,000 is payable until you pay it.

- **Debt**: Borrowed money owed to lenders, usually with interest.

  A $1M bank loan taken out to fund expansion.

- **Equity**: Owners' residual stake — assets minus liabilities.

  Founders put in $100,000 and the company has since kept $400,000 of profit rather than paying it out — equity is $500,000.



1.4.1.1. Operating Working Capital (OWC)

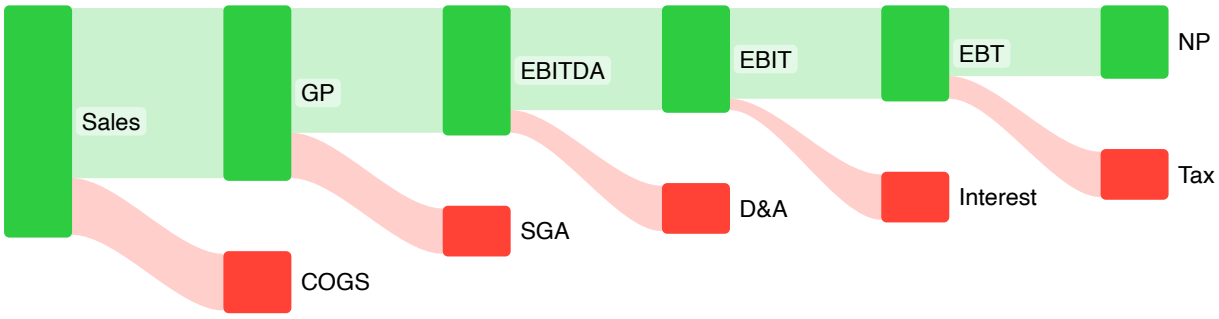
$$\text{OWC} = \text{Min Cash} + \text{Account Receivable} + \text{Inventory} - \text{Account Payable}$$

Assets	Liabilities
Cash	Account Payables
Account Receivables	Debt
Inventory	Equity
PP&E	

$\rightarrow$

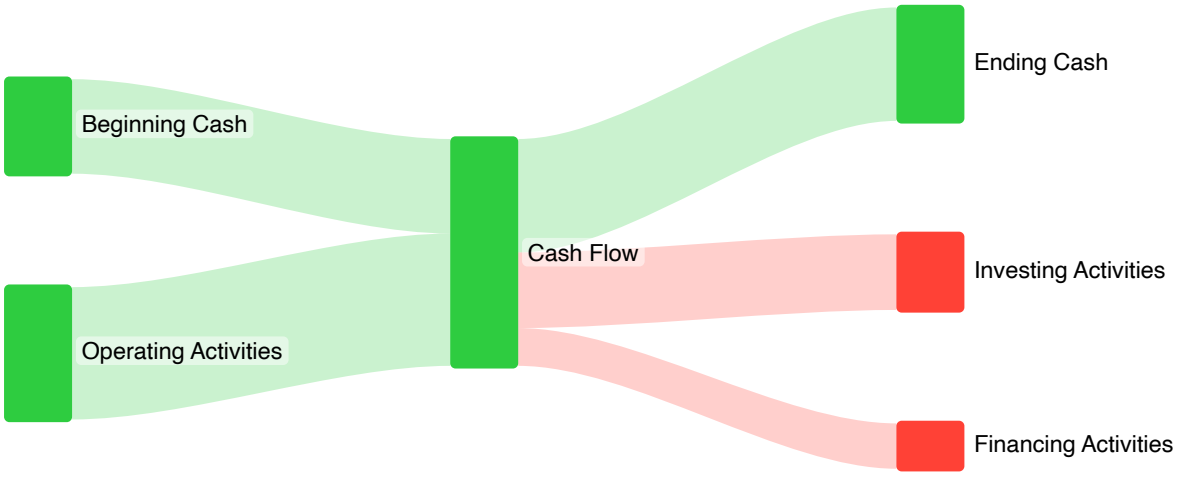
Assets	Liabilities
OWC	Debt
PP&E	Equity

1.4.2. Income Statement (P&L)



- **Sales**: Revenue — total money brought in from selling goods or services, before any costs are subtracted. Also called “revenue” or the “top line.”
- **GP** (Gross Profit): Sales – COGS. What’s left after covering the direct cost of producing or delivering what was sold.
- **EBITDA** (Earnings Before Interest, Taxes, Depreciation & Amortization): GP – SGA in this chain. A rough proxy for core operating profitability, stripped of financing costs, taxes, and non-cash charges — often used to compare operating performance across companies with different debt loads or asset ages.
- **EBIT** (Earnings Before Interest & Taxes): EBITDA – D&A. Also called “Operating Income” — profit from core operations after accounting for the wear-down of assets, but still before financing costs and taxes.
- **EBT** (Earnings Before Tax): EBIT – Interest. Pre-tax profit, after the cost of servicing debt.
- **NP** (Net Profit / Net Income): EBT – Tax. The bottom line — what’s actually left over for shareholders or retained earnings.
- **COGS** (Cost of Goods Sold): The direct costs of producing or acquiring what was sold — materials, direct labor, manufacturing costs. Excludes overhead, marketing, and admin.
- **SGA** (Selling, General & Administrative): Indirect operating costs to run the business that aren’t tied directly to production — sales and marketing, administrative salaries, rent, overhead.
- **D&A** (Depreciation & Amortization): A non-cash expense that spreads the cost of long-lived assets over their useful life — depreciation for tangible assets like PP&E, amortization for intangibles like patents or acquired goodwill.
- **Interest**: Interest expense — the cost of servicing debt, i.e. interest paid on loans and bonds.
- **Tax**: Income tax expense — taxes owed on EBT.

1.4.3. Cash Flow (CF)



Example: Ethan’s New Firm

On January 1, 2023, Ethan Buell and his friend Ryan started a business to import product X from Germany and sell it in the United States. Product X had been launched successfully in Germany the previous year, and Ethan and Ryan believed it offered a sound business opportunity.

Facts from January 1 to June 30, 2023

1. On January 1, Ethan and Ryan each deposited $10,000 in a bank account to start the business.
2. Also on January 1, the firm borrowed $20,000 from a local bank at 10% interest per year. The loan is expected to be repaid in installments: 50% on June 30 and 50% at the end of the year.
3. On the same day, they rented a store for the six-month accounting period and paid the full rent of $12,000 in advance.
4. The firm bought equipment needed to operate the store for $9,000 cash. Ethan and Ryan estimate that the equipment will last three years and will have no residual value.
5. The first batch of product X cost $36,000. The firm paid $15,000 cash and agreed to pay the remainder to the supplier later.
6. Some defective units from the first batch, costing $6,000, were returned to the supplier, who agreed to take them back.
7. On January 1, the firm hired a store assistant. Her salary is $1,800 per month.
8. The store opened on January 3.
9. On April 15, a second batch of product X was purchased for $48,000 on account.
10. Sales during the six-month period totaled $80,000. Customers paid 70% in cash. The cost of the units sold was $36,000.
11. The firm paid $18,000 to the supplier, reducing the balance owed for the first batch.
12. The firm collected $8,000 from customers who had initially bought on credit. The remaining customer balance is expected to be collected after June 30.
13. Utilities for the six-month period totaled $2,400 and were paid in cash.
14. During the six-month period, the store assistant was paid $9,800 cash. The remaining amount will be paid shortly after June 30.
15. No interest was paid to the bank during the accounting period.
16. On June 30, the firm recognized the full six months of rent expense and six months of depreciation on the equipment.
17. On June 30, the firm repaid half of the bank loan principal.
18. On June 30, Ethan received a call from an important retailer that was interested in buying a large amount of product X regularly.

Statement of Balance	
Assets	Liabilities & Equity
Cash: (1) 20 | 12 (3) Prepaid: (3) 12 | 12 (16)	Equity: | 20 (1) Bank Debt: (17) 10 | 20 (2)

(2) 20	9 (4)
(10) 56	15 (5)
(12) 8	18 (11)
	2.4 (13)
	9.8 (14)
	10 (17)

A/R	
(10) 24	8 (12)

Fixed Assets	
(4) 9	1.5 (16)

Inventory	
(5) 36	6 (6)
(9) 48	36 (10)

A/P	
(6) 6	21 (5)
(11) 18	48 (9)

Salaries (Accrued not payed)	
	1 (14)

Interest (Accrued not payed)	
	1 (15)

Expenses		Revenue	
(10) CoGS	36	(10)	80
(13) SGA (utilities)	2.4		
(14) SGA (salaries)	10.8		
(15) Interest	1		
(16) Rent	12		
(16) Depreciation	1.5		

Statement of Balance	
Assets	Liabilities
Cash **27.8**	A/P **47**
A/R **16**	Debt **10**
Inventory **42**	Equity **36.3**
PP&E **7.5**	

Income Statement	
Revenue	80
GoGS	**36**
Gross Profit	44
SGA	**25.2**
EBITDA	18.8
D&A	**1.5**
EBIT	17.3
Interest	**1**

EBT	16.3
Tax	0
NP	16.3

Statement of Cash FLow		
Operating	Net Profit	
	D&A	
	A/R	
	Inventory	
	A/P	
	Cash generated by Operations	
Investing	New Investment	
	Cash used in Investement	
Financing	Cash raised from Equity	
	Chnages in Debt	
	Cash used in financing	
	Net cash	

Example: Vending Machines

During his stay in Switzerland in 2012, when he was about to finish his MBA, Sanjay Menon had an idea in mind to start a new business back home in India. In several European airports, he had seen some vending machines selling electronics (like iPods) rather than soft drinks or other inexpensive products. He thought that importing and selling those new-generation vending machines in India could be a good idea, especially when some of his relatives were already involved in the industry of vending services. However, Sanjay did not have much money, so he showed his business plan to his rich uncle Anoop Patnaik, so as to get him involved in the new firm as the main shareholder.

- Anoop accepted to invest in the business, so they both went ahead and made a bank deposit of $120,000 (Sanjay invested $20,000, his uncle $100,000) as equity for the new firm.
- Additionally, at the end of the year, they borrowed some money from a bank ($40,000, 10% p.a.) to avoid potential cash tensions.
- On December 30, they bought a small warehouse to store the machines, whose price was $100,000 ($40,000 the land, $60,000 the building). Sanjay paid $80,000 cash; the remaining amount was agreed to be paid on Dec 30 the following year. Both shareholders estimated the life of the warehouse to be 20 additional years.
- The same day, Anoop bought some necessary furniture to run the business, paying $4,000 cash. That furniture was supposed to last five years.

The last day of the year, Sanjay and Anoop met to assess the situation of the firm. No machines had been sold yet. Sanjay had prepared the cash report shown below.

- Sanjay also mentioned that there were 20 machines in the warehouse, whose total cost had been $2,000 each. Anoop read the report and told Sanjay that the information was not sufficient. He then asked for the Balance Sheet of the firm (Dec 31, 2012). Sanjay did not know exactly how to prepare it, but did his best.

During 2013, Sanjay was in charge of the business, taking care of cash flows (i.e., cash inputs and outputs), but not really paying attention to the financial situation of the firm. Time went by quickly. At the beginning of December 2013, Sanjay got an e-mail from his uncle requiring the following information for year 2013:

- Balance Sheet (Dec 31)
- Income Statement
- Statement of Cash Flows

Sanjay gathered all the information he was able to find regarding operations during 2013 and summarized it as follows:

1. The firm imported 170 additional machines (on account, i.e., using trade credit), $2,000 each.
2. Fifty machines were sold (cash), at $3,000 each.
3. One hundred machines were sold (on account), $3,300 each.
4. During 2013, $250,000 were collected from customers who had not paid cash.
5. Likewise, the firm paid the cost of 140 machines to the supplier.

6. General and administrative expenses totaled $20,000 (paid cash).
7. The remaining money corresponding to the warehouse was paid.
8. On Dec 31, 2013, half of the bank debt (i.e., $20,000) was repaid.
9. Interest of the loan was paid.

Statement of Balance	
Assets	Liabilities & Equity
Cash PP&D	A/R PP&D	Equity	A/P

Expenses	Revenue

Statement of Balance	
Assets	Liabilities
Cash 0	A/P 0
A/R 0	Debt 0
Inventory 0	Equity 0
PP&E 0	

Inventory valuation

1. Product costing
    - Direct v. Indicrect Costs
        - Direct: Expenses that directly go into producing goods specific to the operation (direct labor, direct materials, mfg supplies)
        - Indirect: General business expenses for operations shared between operations (indirect labor, rent, utilities)
    - Fixed v. Variable Costs
        - Fixed: Does not depend on the volume of activity
        - Variable: Varies in relation to volume
2. Flow assumption
    - Absorbtion costing
        - Direct cost

        - Fixed & variable overhead (indirect costs)
- Direct (variable) costing
- Activity based costing